

Software Requirement Specification (SRS)

1. Introduction

1.1 Purpose

This document outlines the software requirements for an e-commerce platform with categories: Clothing, Food, and Electronics. Features include authentication (JWT, Google OAuth, WhatsApp OTP) using NestJS, Prisma, PostgreSQL, Docker, and Next.js frontend with ShadCN.

1.2 Scope

- Multiple auth options
- Browse categorized products
- User profile & orders
- Admin dashboard
- Containerized deployment

2. System Architecture

Tech Stack:

Frontend: Next.js, ShadCN

Backend: NestJS, Prisma

Database: PostgreSQL

Auth: JWT, Google OAuth, WhatsApp OTP

DevOps: Docker

3. Functional Requirements

User:

- Register/Login via Email, Google, WhatsApp
- View/Edit profile, Order history

Admin:

- Manage products/users
- Role-based auth

Product:

- CRUD, categories, filters

Order:

- Cart, order placement, view history

4. Non-Functional Requirements

Security: JWT, validation

Performance: <1 sec API response

Scalability: Dockerized modules

Usability: Clean responsive UI

Maintainability: Modular structure

Deployability: Docker support

5. Database Schema (Simplified)

Users: id, fullName, email, phone, role, createdAt

Products: id, name, price, category, stock

Orders: id, userId, status, totalAmount

OrderItems: id, orderId, productId, quantity

6. API Endpoints (Sample)

POST /auth/register

POST /auth/login

GET /products

POST /orders

7. Deployment Strategy

Docker for NestJS, DB, Prisma

Frontend on Vercel

Backend on Render/Railway/EC2

8. Future Enhancements

- Payment Integration
- Admin Analytics
- Reviews & Ratings
- Mobile App Support